SPRING END SEMESTER EXAMINATION-2024

6th Semester B.Tech

NATURAL LANGUAGE PROCESSING

IT3035

(For 2021 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

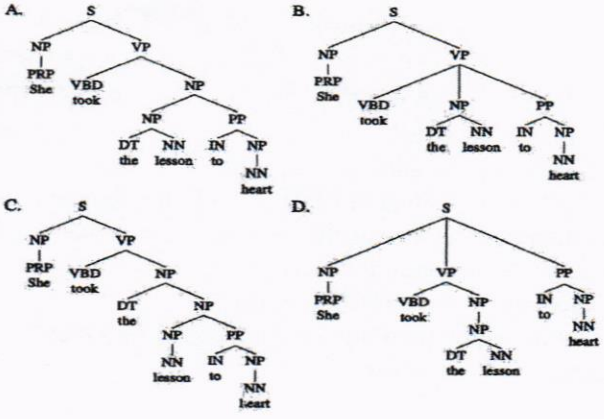
1. Answer the following questions: [1 × 10]

(a) Suppose you have the following training data for Naive Bayes:

I liked the movie [LABEL=+]
I hated the movie because it was an action movie [LABEL=-]
Really cool movie [LABEL=+]

Suppose we are given an unseen input sentence the movie. What is the joint probability P(−, the movie)?

(b) Indicate the correct tree for the most natural interpretation of the following English sentence:



(c) Fill in the following table about algorithms discussed in the NLP course. Give the best single answer for each cell by filling in the blank or circling a value (for the Model Type column). Possible values for the blank cells are given below. You can use the acronyms for SGD and MLE. Options: SGD (stochastic gradient decent), tagging, MLE (maximum likelihood estimation), classification

Method	Task Type	Model Type	Training
Logistic regression		Generative / Discriminative	
Naive Bayes		Generative / Discriminative	
HMMs		Generative / Discriminative	
Feedforward networks		Generative / Discriminative	

(d) Suppose we have these trees:



What grammar would be estimated from this? Write down the rules and their probabilities. You don't need to write down the nonterminals, terminals, or the start symbol. (Hint: the nonterminals are VP, VB, NP, and NN, and each nonterminal has at least one rule with that nonterminal on the left-hand side.)

(e) Suppose you are training a unigram bag-of-words model. Which of the following preprocessing techniques will reduce the size of your feature space? List out all that apply:

I. Lowercasing the data
II. Using bigram features instead of unigram features.
III. Stemming (removing suffixes from words like -ed from verbs or -s from plural nouns)
IV. Removing stopwords (the, of, etc.)
V. Tokenizing punctuation (splitting great! into great!)
VI. Using tf-idf weighting

(f) In an HMM, the tag transition probabilities can be interpreted as:
   A. The probability of seeing a word given a POS tag.
   B. The probability of seeing a POS tag given a word.
   C. The probability of seeing a POS tag given all prior tags, not conditioned on the words.
   D. The probability of seeing a POS tag given the preceding tag, not conditioned on the words.
   E. The probability of seeing the next POS tag given the current word

(g) Define Perplexity measure in NLP.

(h) Explain the benefits of eliminating stop words. Give examples for situations in which elimination of stop words may be harmful.

(i) Write the differences between top-down and bottom-up parsing methods.

(j) Consider a corpus of English containing approximately 50,000 tokens. In this corpus we have the counts of unigrams and bigrams as per the table below. Estimate P(good) and P(good|boy) using maximum likelihood estimation without smoothing.

good	boy	boy good	girl	girl boy
3,700	1,200	65	1,300	0

SECTION-B

2. (a) Consider a simple HMM POS tagger with only five tags (plus the beginning and end of sentence markers, <s> and </s>). The transition probabilities for this HMM are given by the table on the left below, where cell [*i*,*j*] is the probability of transitioning from state *i* to *j* (i.e., $P(\text{state}_j|\text{state}_i)$). A subset of the output probabilities are given by the table on the right, where cell [*i*,*j*] is the probability of state *i* outputting word *j* (i.e., $P(\text{word}_j|\text{state}_i)$). We assume there are other possible output words not shown in the table, and that the <s> and </s> states output <s> and </s> words, respectively, with probability 1. [5]

	CD	PRP	NN	VB	VBD	</s>
<s>	.5	.2	0	.3	0	0
CD	.2	0	.3	.2	.2	.1
PRP	.1	.1	0	.3	.4	.1
NN	.05	.15	.2	.25	.3	.05
VB	0	.2	.6	0	0	.2
VBD	0	.1	.6	0	0	.3

	one	cat	dog	bit	...
CD	.1	0	0	0	
PRP	.02	0	0	0	
NN	.05	.03	.04	.007	
VB	0	0	.03	0	
VBD	0	0	0	.06	

Using the HMM probability matrices, compute $P(w, q)$ (the joint probability of words and tags) for the sentence w = <s> one dog bit </s> with tags q = <s> CD NN NN</s>.

(b) With reference to Question 2 (a), hand-simulate the Viterbi algorithm in order to compute *highest probability* tag se- quence q' for the given sentence, and the joint probability $P(q', w)$, without enumerating all possible tag sequences. That is, fill in the cells in the following table, where cell $[j, t]$ should contain the Viterbi value for state j at time t, and you should also use backpointers to keep track of the best path. The rows of the table are already labeled with the different states, and the columns are already labeled with the observations at each time step. [5]

	<s>	one	dog	bit	</s>
<s>					
CD					
PRP					
NN					
VB					
VBD					
</s>					

Hint: For this particular HMM, a lot of the cells will have zeros in them. Try to work out ahead of time which these are, so you only need to do the Viterbi computations for the other cells.

3. (a) Explain the challenges in creating annotations of training data for NLP applications. What are some of the factors that one should consider? Write down some of the popular evaluation metrics used in NLP applications. [5]

(b) Briefly explain the similarities and differences between the following four types of NLP algorithms: [5]

- N-Gram model
- Naïve Bayes
- Hidden Markov Model
- Context free grammar

4. (a) Briefly discuss some of the challenges in machine translation. [5]

(b) Explain the CKY parsing algorithm and the steps involved in the same. Below is the list of rules in the Grammar: [5]

S → NP VP
NP → D N | Pro | PropN
D → PosPro | Art | NP 's
VP → Vi | Vt NP | Vp NP VP
Pro → i | we | you | he | she | him | her
PosPro → my | our | your | his | her
PropN → Robin | Jo
Art → a | an | the
N → cat | dog | duck | saw | park | telescope | bench
Vi → sleep | run | duck
Vt → eat | break | see | saw
Vp → see | saw | heard

For an input sequence $_0$ **he**$_1$ **saw**$_2$ **her**$_3$ **duck**$_4$, the initial chart of the CKY algorithm looks like this:

	1	2	3	4
0	Pro			
1		Vt,Vp,N		
2			Pro, PosPro	
3				N,Vi
	$_0$he$_1$	$_1$saw$_2$	$_2$her$_3$	$_3$duck$_4$

Apply the subsequent CKY steps and fill the relevant cells (i, j) to reach to the **S.**

5. (a) Consider the following grammar: [5]

DT → the 1.0
NN → cat 0.5
NN → rat 0.5
VBD → chased 0.25
VBD → attacked 0.25
VBD → scared 0.25
VBD → ate 0.25
S → NP VBD 1.0
NP → DT NN 0.5
NP → NP S 0.5

What is the parse of ***the rat the cat chased ate***? Hint: there is only one valid parse of that sentence under this grammar. (Note that this is actually a grammatical English sentence; read it as the rat that the cat chased ate, so a rat is eating something, with that removed to form a reduced relative clause.)

(b) Explain N-gram language models. Discuss different types of N-gram models with examples. List out the problems associated with N-gram models and how these can be handled. [5]

6. (a) Let's see an example of the probabilistic CKY chart, using the grammar which is already in CNF: [5]

S → NP VP	.80
NP → Det N	.30
VP → V NP	.20
Det → the	.50
Det → a	.40
N → meal	.01
N → flight	.02
V → includes	.05

Given this grammar, the below shows the first steps in the *probabilistic CKY* parse of this sentence: ***The flight includes a meal.***

The beginning of the probabilistic CKY matrix is given below. Fill out the rest of the chart. Ignore the grey blocks.

Det:0.40 [0,1]	NP: 0.30*0.40*0.02 = 0.0024 [0,2]	[0,3]	[0,4]	[0,5]
	N:0.02 [1,2]	[1,3]	[1,4]	[1,5]
		V:0.05 [2,3]	[2,4]	[2,5]
			[3,4]	[3,5]
				[4,5]

The flight includes a meal

(b) Explain the Noisy Channel Model. What kind of NLP problems can be solved using Noisy Channel Model? How is a Noisy Channel Model used for Spelling Correction? Explain with a suitable example. [5]

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